Nitrogen pollution reduction in Lee County surface water and groundwater through the replacement of legacy conventional septic systems with aerobic treatment units (ATU)

Luke R. Roides
FGCU Graduate Student
U.A Whitaker College of Civil Engineering, Florida Gulf Coast University, Fort Myers, FL, USA
Email: lroides0791@eagle.fgcu.edu
Phone: 585-698-7236

Abstract

Lee County has experienced urbanization dating back to the 1950s. Residential construction at the time had little to no regard for the future of what Lee County was going to become. Many of the wastewater treatment systems that were constructed for residential properties, known as septic systems, were installed with little to no regulatory enforcement or environmental considerations. Many of these systems are still in use today. With the county's population continuing to grow each year, the surface and ground waters are at risk of contamination from older septic systems. A high number of these septic systems cannot meet the treatment requirements of today's standards and lack separation from the ground water table. Harmful pollutants found in human wastewater, like nitrogen, can cause several environmental and public health impacts. The study objective is to project the reduction of these pollutants from replacing outdated septic systems with new advanced treatment unit septic systems known as aerobic treatment units (ATUs). To explore the reduction capabilities, a mass balance approach and spatial analysis were used to estimate nitrogen loading from legacy conventional septic system. A groundwater flow and nitrogen transport model were simulated in ArcGIS Pro using ArcNLET. Two hypothetical scenarios were modeled to depict the current septic effluent conditions and ATU conversion conditions. The nitrogen reduction findings from this study are not only beneficial for the environment but for public health as well.

Impact Statement

Historically, residential wastewater treatment was not a huge environmental concern during early development. This study highlights the importance of proper treatment and its impact on environmental and public health. The results may strengthen the justification of outdated septic systems needing to be replaced in older densely populated communities to improve water quality throughout the county.

1. Introduction

Nutrient pollution has continuously been an environmental challenge in regions of Southwest Florida. Lee County, which has experienced rapid urbanization over many decades, has been faced with several water quality challenges. The nutrient pollutant impairments of waterbodies are primarily from nitrogen (N) and phosphorus (P) and have been a growing concern not only in Lee County, but also the United States (Diaz-Elsayed et al., 2017). Many coastal ecosystems' primary nutrient pollutant responsible for water quality impairment is nitrogen. Onsite sewage treatment and disposal systems (OSTDS) have been found to be major sources of nitrogen pollution (Diaz-Elsayed et al., 2017). Approximately 30% of Florida's population rely on septic systems (Florida Department of Environmental Protection, 2025). Lee County, FL experienced an urbanization boom throughout the 1950's where conventional septic systems were installed with little to no regulatory requirements. A conventional septic system is typically comprised of a septic tank and a drain field (Diaz-Elsayed et al., 2017).

Costal urban development without centralized sewer systems has led to residential septic systems surrounding canals and estuaries. The Increasing nutrient pollutants such nitrogen making it into coastal waters has led to accelerating eutrophication and promoting harmful algal blooms (HABs) (Brewton et al., 2022; Herren et al., 2021). These anthropogenic impacts can result in habitat destruction, fish kills and impact human health (Brewton et al., 2022). While conventional septic tanks are not highly efficient at the removal of nitrogen, drainfields have been found to be capable of removing 10-40% of nitrogen according to the US environmental Protection Agency. Maintaining nutrient removal levels is essential for the sustainability of human health and aquatic ecosystems (Hossain et al., 2010)**.** Conventional septic tanks do not promote nitrification due to the lack of an aerobic environment. Toxic compounds like nitrate and nitrite can pollute the environment if not properly removed. Both nitrate and nitrite can cause numerous human health problems. Methemoglobinemia (MHB) is a condition that can affect infants leading to cyanosis, also known as blue baby syndrome. Also, nitrate can form N-nitroso compounds in the human digestive system and is considered a carcinogen (Hossain et al., 2010). There are many potential impacts from OSTDS especially in areas where systems are densely populated and within a coastal environment like Lee County.

Lee Countys population has been growing exponentially since the 1950s with approximately 24,000 people at the time. The population doubled annually until the 1980s when the population reached 200,000. Today Lee County's population is approximately 850,000 and is continuing to grow (U.S. Census Bureau, 2025). The urbanization was so rapid that many households were built prior to centralized sewer systems. This resulted in septic systems being the main form of wastewater treatment. At the time there were little to no regulatory or environmental requirements regarding the installation and maintenance of septic systems. Many of these systems were installed with no consideration of the environmental or public health impacts associated with the pollutants form wastewater. It was not until 1983 when the

Florida Administrative Code (FAC) implied more strict constraints on septic system installations, even then systems continued to be installed freely throughout the 1980's and 1990's. This is critical to Lee Countys surface water and groundwater quality due to many of these legacy systems still being in use today. Septic drainfields installed with no consideration of surface and groundwater are actively contaminating the local waters with harmful nutrients like nitrogen. Effluent from septic system has vertical and lateral boundaries that can exceed hundreds of meters and infiltrate groundwaters (Subedi et al., 2015). With ground water table separation from septic drainfields not being enforced until the late 1990's – early 2000's, and Lee County having a high groundwater table many of these systems do not have the separation to obtain any of the nitrogen removal capabilities in the drain field (Florida Department of Health, 2025). While there have been previous studies analyzing enhanced nutrient removal from septic systems, there are no local studies in Lee County that analyze the potential nitrogen reductions through the replacement of older conventional systems with advanced septic systems like an Aerobic Treatment Unit (ATU). ATUs can achieve 50% or more total nitrogen (TN) reductions through biological nutrient removal (Diaz-Elsayed et al., 2017). Historical data estimates that 10 – 20% of septic systems malfunction or go into failure annually (Du et al., 2014). Septic system failure from age or hydraulic overload can result in sewage effluent ponding onto the ground surface leading to an increase in pollutant surface runoff (Withers et al., 2014). The replacement of the older and potentially failing septic systems in Lee County with new regulatory requirements and advanced ATU systems has the potential to be a beneficial alternative for onsite wastewater treatment in areas with no sewer access. By evaluating the replacement of outdated septic system, this project measures the potential environmental and human health benefits from implementing ATUs to reduce nitrogen from contaminating surface water and groundwater in Lee County.
This leads to our research questions which include:

- Will replacing old septic systems with advanced aerobic treatment units in Lee County reduce nitrogen loading to surface and ground waters?
- How will current water quality trends compare to ATU replacement scenarios?

The project is a significant topic due to Lee County experiencing continued water quality issues and rapid urbanization. The project focuses on Lee County, as it is a historical city that has greatly expanded over many decades and has been faced with rapid urbanization and water quality issues. Focusing on Nitrogen is reasonable as it is one of the most impactful pollutants of concern coming from septic systems. There have been extensive studies on conventional and ATU septic systems which has provided the study with reliable existing data. We have access to data on all existing septic systems in Lee County through the Florida Department of Health. The project will focus on rich nitrogen hot spots specifically surrounded by densely urbanized residential areas serviced by older septic systems. Lee Countys public water quality sampling

data allows for easy analysis of watershed water quality. Putting a focus on specific nitrogen hot spots and utilizing available resources has made the study significant for options of improving water quality in areas of need.

2. Method

This study explores a quantitative modeling approach utilizing existing data.
While the research questions will be addressed throughout this study, the future of onsite wastewater treatment and pollutant reductions may be considered for Lee County environmental and public health.

2.1 Study Site.

Lee County was chosen for the study due to the dense coastal population and historical water quality issues that have been observed throughout local waters, like the Caloosahatchee River (CHNEP, 2025; Tyre et al., 2023). The Stroud Creek watershed in North Fort Myers was chosen as the priority study site within Lee County due to the minimal sewer availability and historical septic systems.

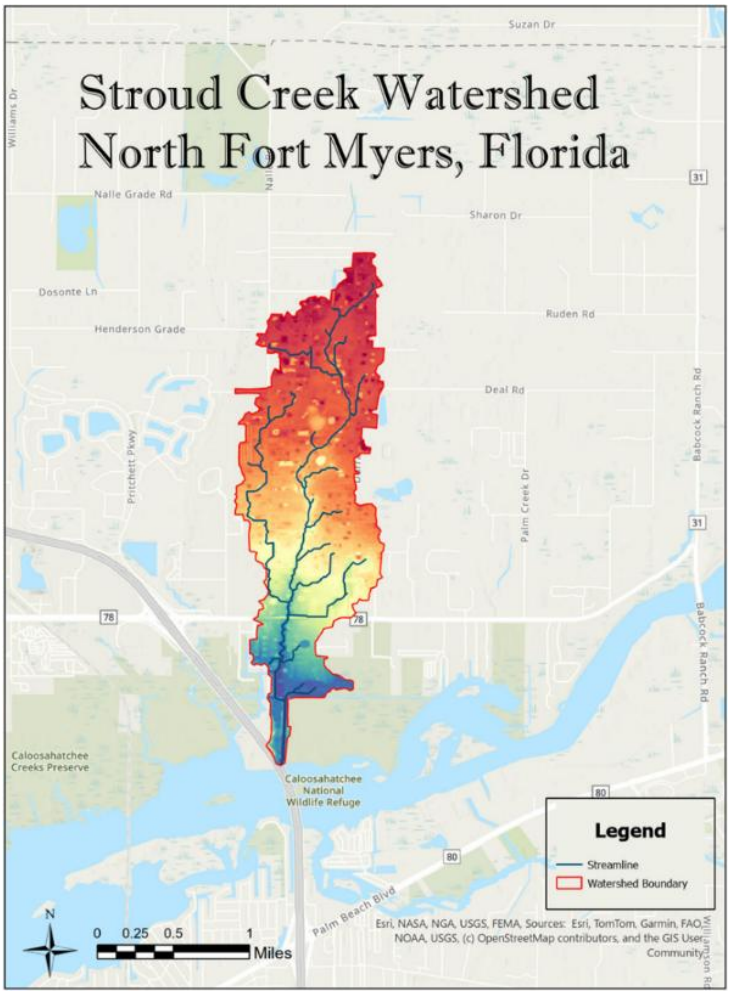

Figure I Study Area: Stroud Creek delineated watershed boundary

Many historical septic systems in the North Fort Myers region were installed with little to no regulatory enforcement or environmental considerations and are still in use today. Today many of these systems remain installed directly contaminating ground waters making it an ideal study site for modeling the reductions in TN from replacement ATUs (Buszka & Reeves, 2021).

2.2 Research scenario design

Two hypothetical scenarios were evaluated using a spatially distributed groundwater modeling approach to determine the impact of replacing legacy septic systems with advanced septic systems to assess nitrogen removal capabilities to surface water and groundwaters. The two scenarios developed had consistent system characteristics as far as flow and system sizing. For the purpose of this study, legacy conventional septic systems for each household are considered to have a 900-gallon septic tank and an aggregate drain field.

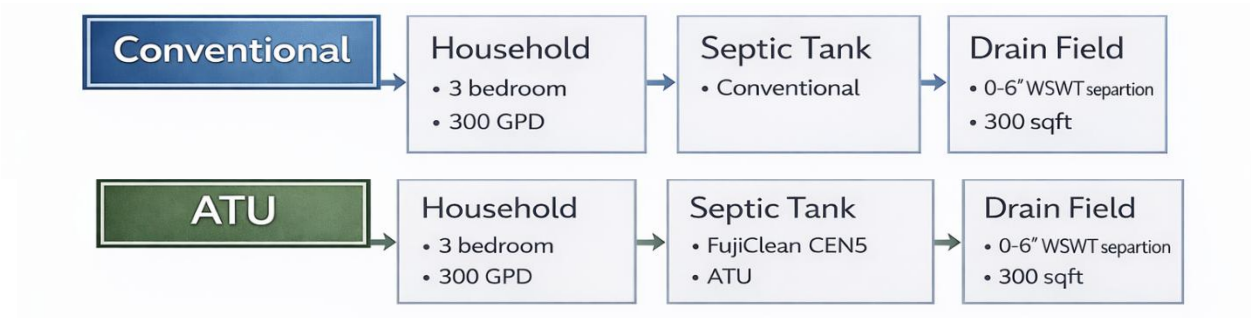

Figure II Hypothetical scenarios for pre and post septic system conversion

According to the Florida Administrative Code (F.A.C) septic systems design criteria considers number of bedrooms for estimated sewage flow, assuming 2 persons per bedroom, and each person producing 50 gallons of wastewater per day (FAC 62-6, 2025; Buszka & Reeves, 2021). A wet season water table (WSWT) separation for the drainfield was assumed to be 0 to 6 inches to replicate the conditions of most legacy drainfield separations found in the high WSWT areas of Lee County like North Fort Myers. Today's separation requirements would require to be 24" from the WSWT (FDOH, 2026). Scenario 2 would consist of the same flow and WSWT separation but would replace the conventional tank with an aerobic treatment unit, specifically the FUJI CLEAN CEN5. Drainfield sizing remained constant across the scenarios to replicate smaller sized drainfields typical for legacy systems of 300 square feet, today's requirements for a 3-bedroom household would require 500 square feet. With household-to-household numbers of exact occupants being difficult to determine, scenarios will be considered to have a 300 gallon per day (GPD) estimated sewage flow which is typical for a 3-bedroom household to isolate the effects of nitrogen reductions between the two systems (U.S. Census, 2025).

2.3 Data sources and preparation.

Spatial datasets were processed using ArcGIS Pro. The Stroud Creek watershed was delineated from a digital elevation model (DEM) and hydrologic analysis tools, including flow direction, flow accumulation, and fill to generate a watershed boundary that was used to clip all spatial datasets for the study area. Surface water features and the discharge point for the watershed were used to create the Stroud Creek streamlines. All spatial data were projected to NAD 1983 UTM Zone 17N to remain consistent across all inputs. Septic system location points were approximated using residential parcel data along with Florida Department of Health records (LEEPA, FDOH, 2026).

2.4 Nitrogen Loading Parameters

Septic system effluent nitrogen concentrations assigned to each individual system were based on published literature and regulatory values of similar systems. Conventional septic system effluent concentrations were chosen to be 72mg/L total nitrogen (TN). This value was based on a study conducted by Humphrey et al., 2013 where they had sampled two similar sized systems monthly from 2009 to 2010 that matched characteristics as systems modeled in scenario 1. Scenario 2, which replaces the conventional tank with the FUJI CLEAN CEN5 ATU, has been NSF245 certified to achieve 70-75% reductions in TN. Third party state required sampling has proven the system is capable of achieving a 90% TN removal rate. Scenario 2 conditions were based on a best case scenario level of 90% reduction in TN (Fuji Clean USA, 2025; Gobler et al., 2021; Humphrey et al., 2013; NSF, 2025).

2.5 Transport Modeling

Nitrogen transport was simulated using ArcNLET, a GIS based model that can simulate nitrogen transport from septic systems using groundwater flow and particle tracking parameters (Ye et al., 2015). ArcNLET was selected due to the ability to simulate spatially distributed nitrogen transportation at a watershed scale. Soil data for the Stroud Creek watershed was obtained through the United States Department of Agriculture Web Soil Survey and was used to define groundwater flow. These data and the preprocessing watershed steps were used to generate flow direction and groundwater velocity within ArcNLET (Natural Resources Conservation Service, 2026).

Septic systems points from the preprocessing stage represented the point sources of nitrogen input to the groundwater. Particle tracking was then applied to simulate groundwater movement paths from each septic system point. The nitrogen transport was modeled using advection and dispersion processes while the ArcNLET default settings and soils characteristics implement porosity, dispersivity and decay rate. The modeled outputs consist of spatially

distributed septic system nitrogen concentration plumes extending across the entire watershed for 298 septic systems.

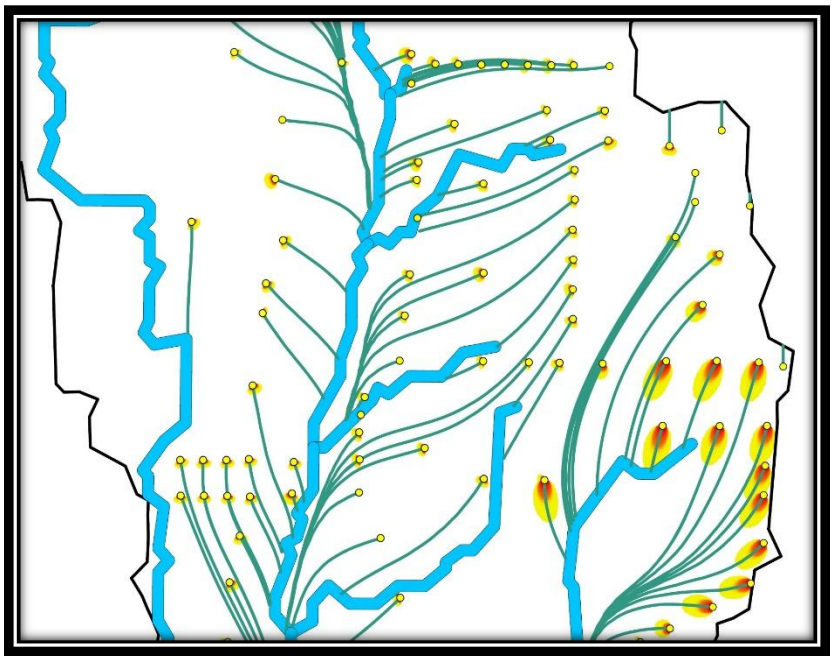

Figure III ArcNLET Particle tracking and plumes for area of interest

2.6 Data Analysis and Statistical Methods.

Modeled nitrogen plume concentrations were exported from ArcNLET as a raster and converted to tabular datasets. The datasets were analyzed using Python to evaluate the statistical and spatial differences between the two scenarios. Histograms were created to analyze frequency distributions of the nitrogen plume concentrations. Cumulative distribution functions (CDFs) were created to evaluate the proportional difference between scenarios as well as assess concentrations exceeding the nitrate (NO_3^-–N) drinking water standard threshold of 10 mg/L (United States Environmental Protection Agency, 2026). The CDF analysis was selected to help provide a distribution based nitrogen comparison between the two systems and provide distinct reduction capabilities reflected in mean values alone throughout the entire plume distribution system model.

2.7 Assumptions, Limitations, and Validity

The study involved assumptions to support the modeled implementations. The groundwater modeling in ArcNLET was treated as a steady state scenario, and septic systems were approximated through parcel data and may not be to scale of existing systems on properties. Nitrogen concentrations were strictly based off previous literature and site specific in field analysis of equivalent systems. No in-field sampling data was derived from septic systems located in Lee County. However, due to modeling structure and system parameters remaining

consistent between two scenarios, the differences shown between system efficiency of conventional and ATUs remains valid.

2.8 Reproducibility

All analyses were conducted using ArcGIS Pro, ArcNLET and Python. All inputs required to reproduce this study consist of DEM layers for flow directions and flow accumulation raster, groundwater and soil parameters for velocity, porosity and flow direction available on the USDA Web Soil Survey website, Contaminant transport parameters are embedded into the ArcNLET program and are fully customizable for specific conditions. Nitrogen concentrations for analysis and septic locations as point sources can be publicly available across a variety of scenarios. With the given parameters ArcNLET modeling can be replicated and generate comparable plume outputs as this study for any desired septic reliant area. Similar statistical analysis can also be evaluated by exporting modeled plume outputs to CSV format and analyzed with available Python Libraries.

3. Results

3.1 Spatial Distribution of Nitrogen Plumes

Modeled ArcNLET groundwater nitrogen plumes for conventional and ATU systems are shown in *Figure IV* and *Figure V* respectively. Under hypothetical existing conditions the nitrogen plumes show much higher concentrations and extend further downgradient from the source.

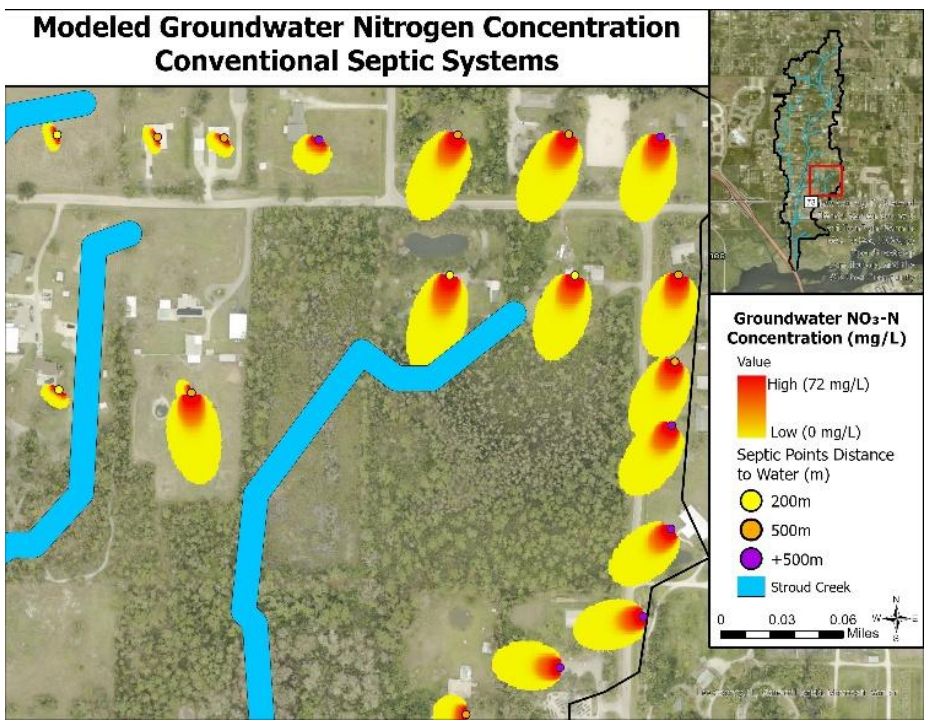

Figure IV Modeled groundwater NO_3-N concentration plumes for conventional septic systems in the Stroud Creek watershed.

Some plumes extending and directly connecting to streamlines of the Stroud Creek watershed indicating a possible connection of contamination between the two.

The ATU scenario shown in *Figure V* shows a clear reduction in levels of nitrogen concentrations across all plumes. Overall plume sizing shows a reduction in comparison to the conventional plumes indicating that the conversion not only reduces levels of nitrogen entering the groundwater but also reduces spatial extent. Transport direction remains the same between scenarios but the decrease in plume size greatly reduces higher levels of nitrogen impeding surface waters.

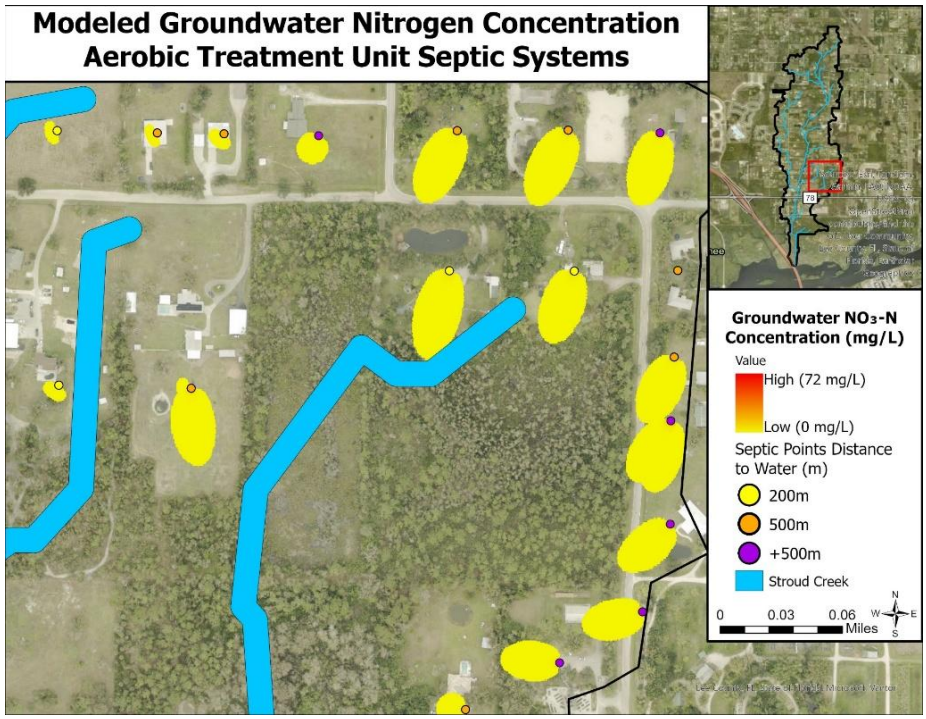

Figure V Modeled groundwater NO_3-N concentration plumes for ATU septic systems in the Stroud Creek watershed.

Figure VI shows the distribution of the nitrogen reduction between the conversion of the systems. Modeled results indicate that reductions are highest at the point source of the septic system where the effluent is discharging into the groundwater. This is significant for the system watershed analysis as a whole due to the ATU discharge levels being greatly reduced at the source causing much lower-level plume extents of nitrogen.

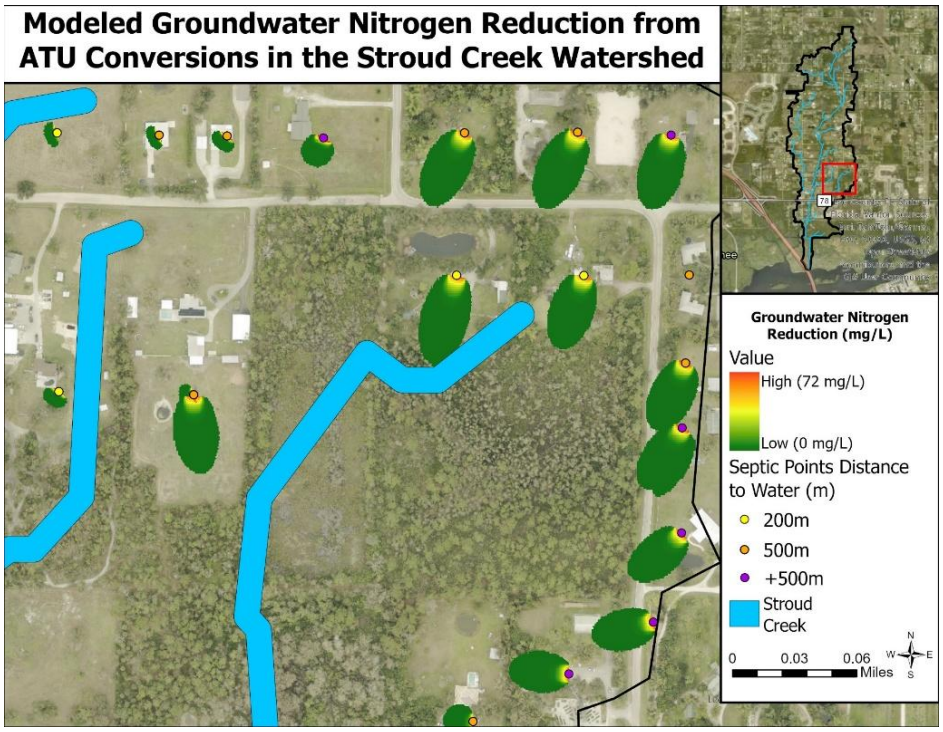

Figure VI Spatial distribution of nitrogen reduction resulting from conventional septic system to ATU conversion. High reductions observed near point sources.

3.2 Distributions of Groundwater Nitrogen Concentrations

The cumulative distribution of NO_3-N concentrations for both conventional and ATU systems can be seen in *Figure VII*.

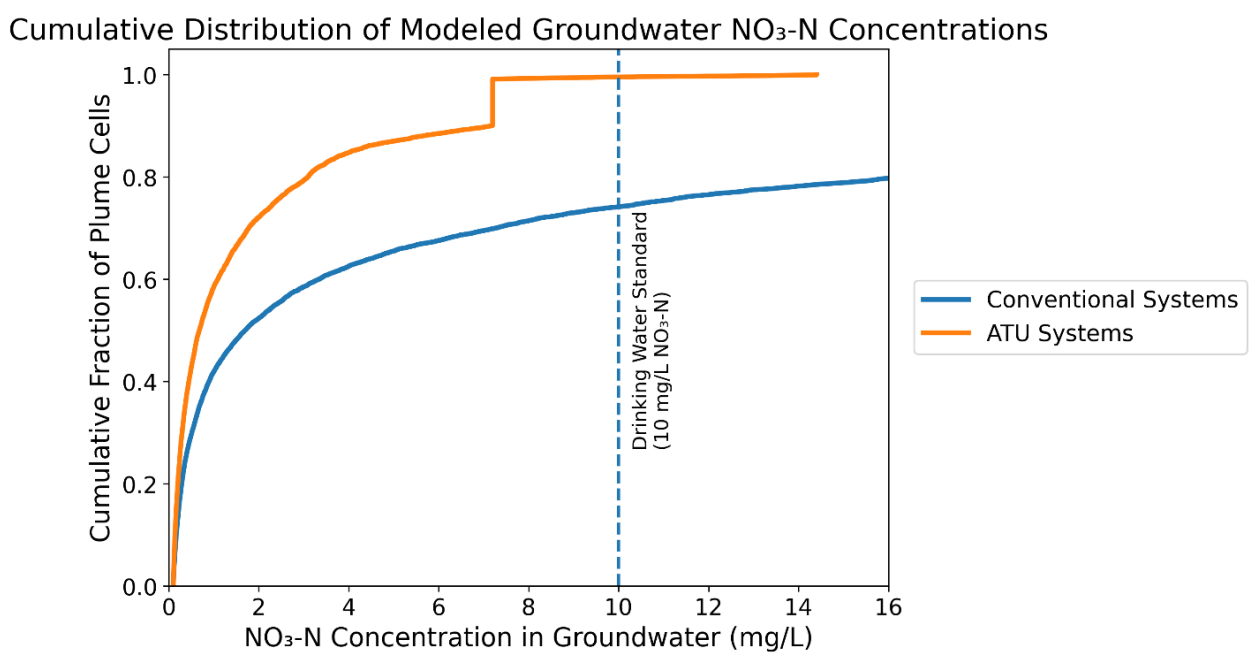

Figure VII Cumulative distribution function (CDF) of modeled groundwater NO_3-N concentrations between both scenarios

The ATU system curve increases rapidly with a steady increase around the 7.2 mg/L mark where the 90% reduction threshold takes place until all plume cells start to plateau with very few exceeding beyond that mark. The conventional system curve has a weaker increase, remaining below the ATU curve entirely, and extending well beyond drinking water standard threshold of 10 mg/L. This reflects that a broader distribution of plume cells reaches much higher levels of NO_3-N concentrations in groundwater than ATU systems.

The distribution of modeled NO_3-N concentrations between systems in *Figure VIII* shows that a majority of cell counts for the ATU systems remain consistently at lower concentrations. While the conventional systems display an extended distribution of much higher concentrations.

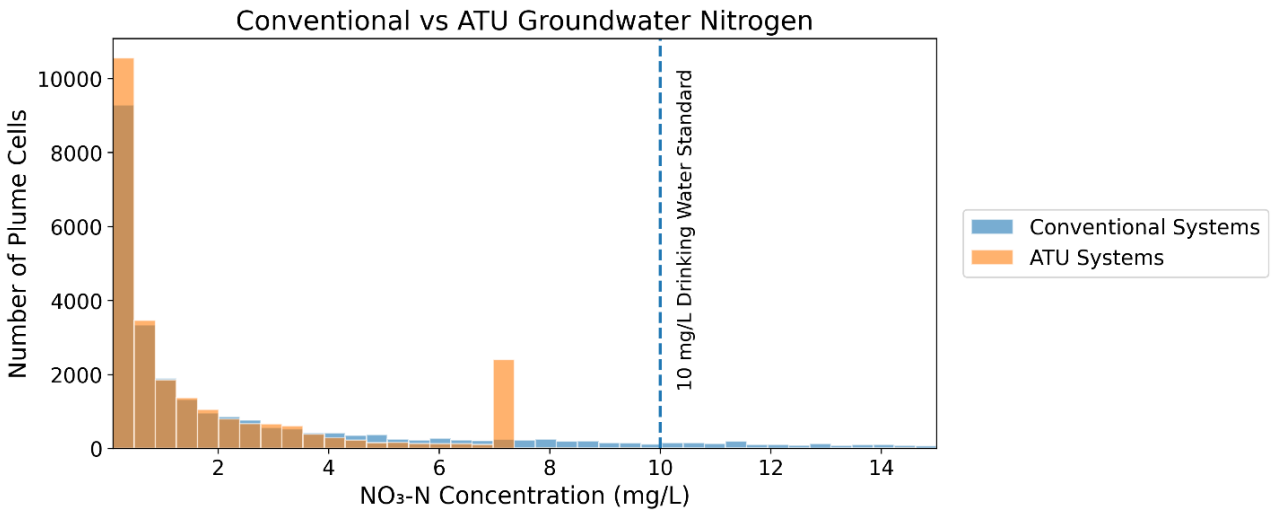

Figure VIII Histogram showing distribution of modeled groundwater NO_3-N concentrations.

The distribution of plume cells across the two scenarios is shown in *Figure IX*.

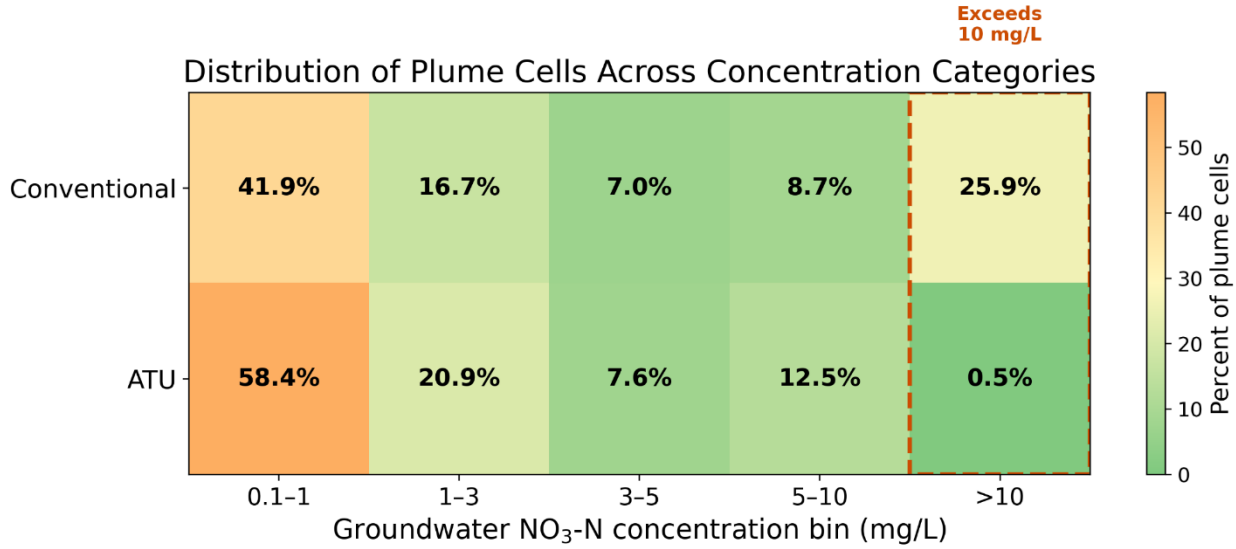

Figure IX Distribution of plume cell concentration between both scenarios

Conventional systems have a majority of their distribution between the 0.1-1 mg/L of NO_3-N concentrations. However, the second highest distribution for conventional systems was found above 10 mg/L of NO_3-N in groundwater at 25.9%. The ATU conversion resulted with a 58.4% of the plume cell concentrations being between 0.1-1 mg/L of NO_3-N, and only 0.5% of cells exceeding greater than 10 mg/L of NO_3-N.

Summary statistics of the modeled concentrations are presented in *Figure X*. Conventional systems exceeded ATU levels entirely. The conventional systems nitrogen plume concentration for the entire watershed had a mean concentration of 11.79 mg/L while the ATU scenario had 1.81 mg/L. The conventional systems had a median concentration of 1.70 mg/L and a 90th percentile of 45.21 mg/L, while the ATU scenario had 0.68 mg/L and 7.17 mg/L respectively. The conventional scenario had a peak concentration of 144.00 mg/L while the ATU scenario reached a maximum level of 14.40 mg/L.

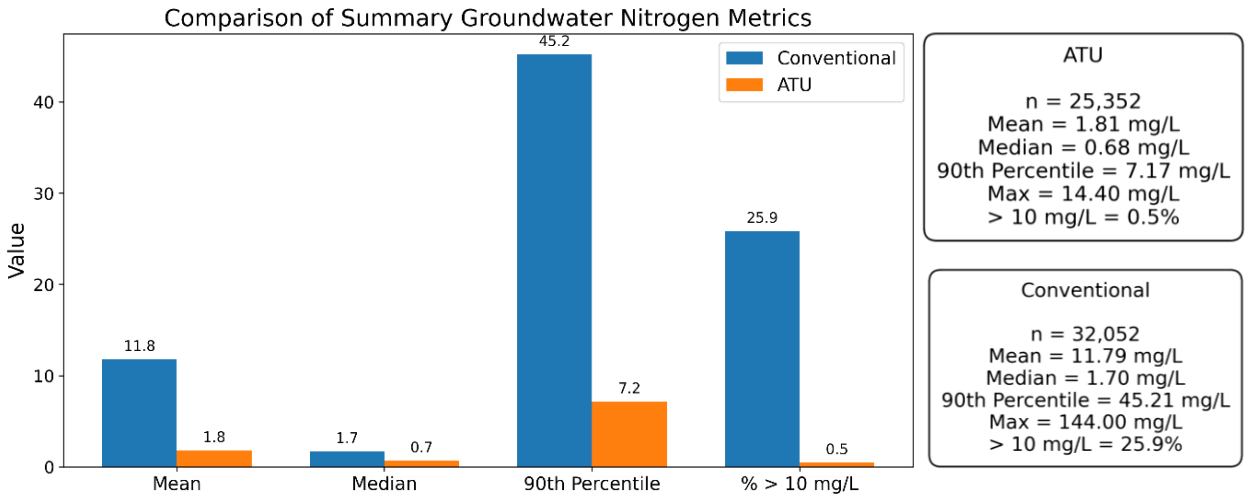

Figure X Statistical summary of groundwater NO_3-N concentrations for conventional and ATU septic systems

These results indicate substantial reductions across upper and lower ranges of nitrogen concentrations between the two scenarios.

4. Discussion

The results of this study demonstrate a clear difference between the two modeled scenarios of legacy conventional septic systems and the aerobic treatment unit (ATU) systems. Across all analysis the ATU had outperformed the conventional system in regard to reductions in nitrogen concentration in groundwater, plume extent, and reduced the entire watersheds plume cells of exceedances of the drinking water standard of 10 mg/L to almost 0%. As the nitrogen reduction observed was to be expected given the literature-based values between the two systems, the magnitude of the reductions across an entire watershed highlighted the potential effectiveness of better water quality with a system wide conversion.

One of the most notable findings of the study was the reduction capabilities in the higher nitrogen level ranges in the ATU model showing a significant decrease in overall levels exceeding 10 mg/L. While the conventional system model not only exceeded upper thresholds of 10 mg/L, over 25% of plume cells in the watershed were impacting groundwater at these extreme levels. The ATU conversion shifted the entire distribution of plume cells to much lower levels with a majority being below 1 mg/L. This shows that the ATU conversion is not only lowering nitrogen levels but also reducing the risk of environmental and public health impacts.

The reduction in plume extent observed between the two scenarios under a spatial analysis concludes that not only are reductions occurring at a concentration level, but ATUs reduce impacts at a spatial level as well. The conventional scenario plumes showed higher levels intersecting with surface water and moving further down gradient from the source. While the ATU conversion scenario not only shortened the plume extent further from surface waters while reducing concertation, it kept higher levels of nitrogen concentration more localized to the point source of the septic system further reducing the risk of nitrogen transport on the surrounding environments.

The findings in this study align with previous studies where septic systems contribute nitrogen pollutants to groundwater and the performance of the ATUs ability to reduce nitrogen has also been well documented (Diaz-Elsayed et al., 2017; Brewton et al., 2022). However, this study extends beyond previous site-specific implication, and the findings show the reduction capabilities at a watershed scale utilizing a modeling approach to replace older septic systems in areas with no sewer availability. Although the reductions were to be expected given the effluent concentrations among the two systems, the extent at which the nitrogen plumes were reduced and cells being decreased to almost completely below the drinking water standard levels show just how critical this septic conversion could be for the environment and public health. Not only for the Stroud Creek watershed but for Lee County as a whole. Upgrading legacy conventional septic systems could serve as an effective strategy for improving surface water and groundwater quality in regions where sewer is not a feasible option.

5. Study Limitations

There are limitations that should be considered when analyzing this study. One of the main limitations of the study is that the ArcNLET modelling was treated as a steady state system with relatively similar subsurface conditions. Realistically groundwater flow can be impacted by seasonal variations over time as rainfall and groundwater levels fluctuate. The model does not capture the natural environmental variability that would occur in the field and should be simply viewed as a general representation rather than exact results.

Another limitation is septic system location, configuration and estimated sewage flow from residences. Septic system locations were estimated with parcel data and may not be exact. Actual sewage flows and septic tank and drainfield sizing can vary from household to household. Although existing data from the Lee County Florida Department of Health indicates no ATUs are currently present in the Stroud Creek watershed, actual performance of conventional systems can vary based on actual age, maintenance and sewage flow. These limitations could result in differences in the actual nitrogen loading compared to the literature-based levels that were modeled.

Lastly, the study assumes a complete conversion of all systems within the watershed when in real world conditions this would be a gradual transition and types of systems and sizing would vary and these factors could potentially affect what nitrogen reductions would actually be achieved. Even with these limitations the modeled scenarios show a meaningful comparison in how the conversion from older conventional systems to ATUs can improve nitrogen loading within an entire watershed.

6. Conclusion

This study covered how replacing legacy conventional septic systems with newer advanced aerobic treatment units could reduce the levels of nitrogen entering the groundwater and impacting the environment and public health. The study found that the conversion to an ATU would consistently produce lower levels of nitrogen at a watershed scale, generate smaller plume extents from every point source, and almost eliminate levels of NO_3-N exceeding the drinking water threshold of 10 mg/L. These findings alone address the research question by showing that upgrading the septic systems in areas where sewer is not feasible can overall improve water quality conditions. This study also shows that reductions are more than just a point based single system result but can improve the water quality of an entire watershed. This is significant for areas like coastal Fort Myres where many areas have a high density of older septic systems and high fluctuating groundwater tables. The improvement in nitrogen reduction from the septic system not only reduces harmful concentrations of nitrogen entering the groundwater but transport modeling also indicated that less surface waters would be impacted as well by nitrogen plumes.

Overall, even though the study is based on a steady state and simplified assumptions, the results still indicate a great reduction of nitrogen pollution can occur from the replacement of older conventional septic systems with advanced onsite wastewater treatment systems like ATUs.

<u>Future perspectives</u>

Future research could focus on using field collected data in the Lee County area to better refine modeling of an entire watershed to reduce uncertainty. It would also be useful to address private well water quality in high density septic areas as many septic dependent areas also rely on private wells servicing each house instead of a public water source. This is important because modeled results indicated that nitrogen plumes cross over property to property and may impact well water quality which can be a major public health risk. In addition to the ATU conversion, a cost evaluation at a watershed level could help determine how feasible large scale septic conversions could cost in comparison to centralized sewer infrastructure.

This research can be used for future policy implementation based on the findings of the benefits of upgrading outdated septic systems. These findings will strengthen the justification of replacing outdated septic systems and help policymakers and stakeholders be more informed for the decision making of future residential wastewater treatment in Lee County. The environmental and public health impacts related to wastewater pollution, such as HABs, may be significantly reduced through the implementation of this study. Furthermore, the public will gain knowledge on the current and future state of residential wastewater treatment.

Data Availability

The datasets and code generated for the study are available in a public GitHub repository available at https://github.com/LukeRR96/TN-Analysis/tree/main, including processed ArcNLET outputs, Python analysis, and other supporting documentation utilized to reproduce results. Publicly available datasets used in the study for parcel data and septic system records can be accessed through the Lee County Property Appraiser (LEEPA) and Florida Department of Health public records. Additional data files are available from the corresponding author upon request.

Ethical Considerations

All data collected for this study is public record for Lee County and no human subjects were involved. No field sampling or interactions with private property were conducted. Data used for this study was backed by peer-reviewed literature sources and government entities. Results are in the form of estimates and not exact future projections.

Acknowledgements

The author would like to acknowledge the guidance and support of faculty at Florida Gulf Coast University, particularly Dr. Ahmed S. Elshall, Dr. Seneshaw Tsegaye, and Dr. Mebrahtom Kebedew for their assistance with project development and technical direction.

AI Assistance Statement

Artificial intelligence tools, including ChatGPT, were used to support the development of python coding. All technical analysis including modeling, data processing and interpretation of results were conducted by the author of the study.

Funding

This study did not receive any formal funding from any specific agencies.

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